







Plasmid Identification, Characterisation, Clusterisation, Interpretation, and Scoring

A unified, high-throughput pipeline for bacterial plasmid analysis.

Macarena Algañaras, Alejandra Bosch, Mauricio Lozano



Plasmids
drive the engines of bacterial evolution



Accelerating
the global spread of
Antimicrobial Resistance (AMR).



Transferring
virulence factors
rapidly across species boundaries.



Generating
evolutionary islands of
polyploidy that defy
chromosomal genetic rules.

The Plasmid Analysis Bottleneck

Current bioinformatic tools offer powerful individual insights but operate in isolated silos.

- Executing multiple tools separately to mitigate bias is **time-consuming.**
- Manually curating conflicting outputs is **error-prone.**
- Integrating functional, taxonomic, and comparative data **lacks standardisation.**



The Unified Solution

piccis

Scoring
A novel, consensus-based reliability metric.

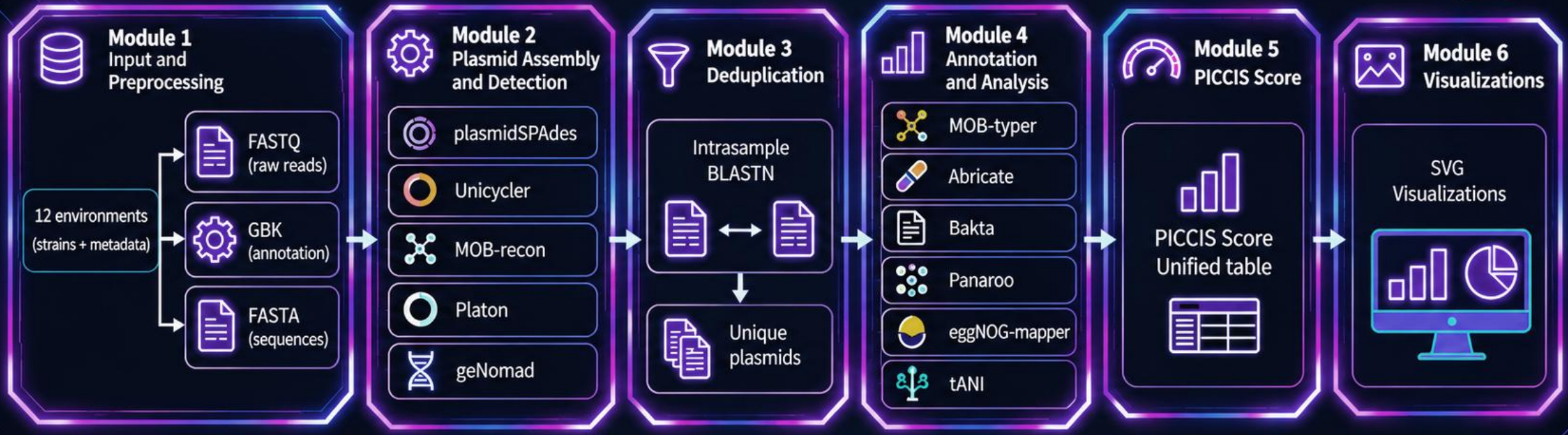
Interpretation
Publication-ready visual synthesis.

Clusterisation
Pangenome construction and population structuring.

Characterisation
Automated functional annotation (AMR, virulence).

Plasmid Identification
Hybrid *de novo* assembly and detection.

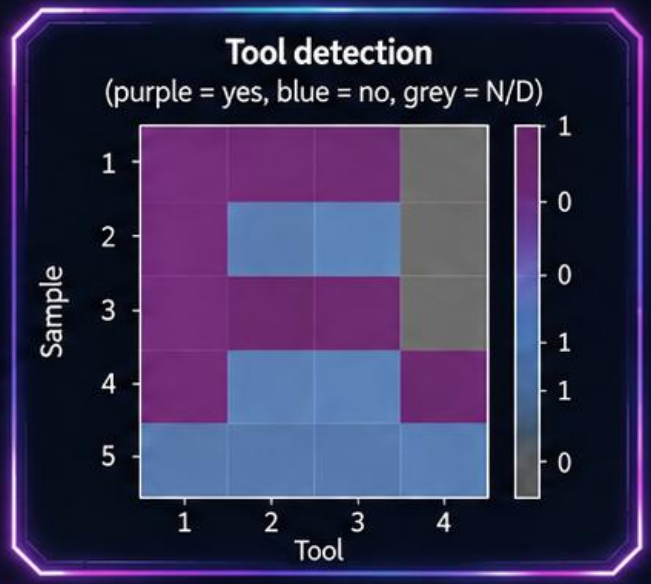
Six-Module Architecture

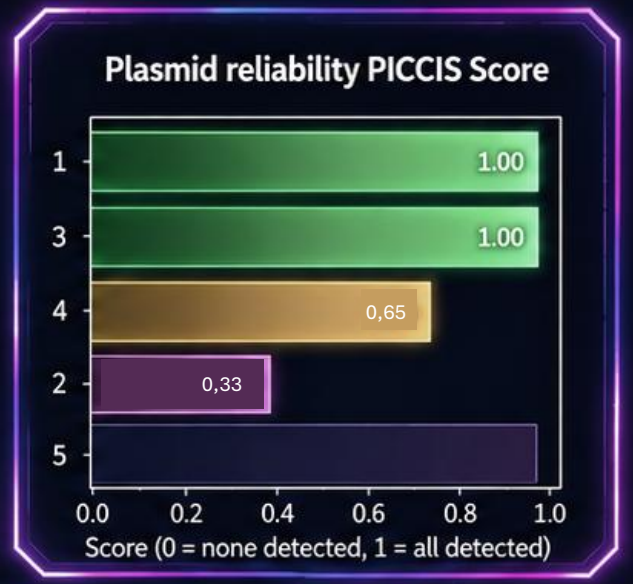


The PICCIS Score: Consensus-Based Reliability

No single tool achieves perfect sensitivity and specificity. PICCIS acts as a panel of independent judges.







Mechanism: SPAdes, MOB-recon, geNomad, Platon, and PlasmidFinder cast independent votes.

Result: A robust metric differentiating high-confidence plasmids from likely chromosomal artefacts, prioritising candidates for experimental validation.

ID	SIZE (bp)	GC (%)	REP_TYPE	MOB_TYPE	PLASMIDFINDER	PICCIS SCORE	CONFIDENCE
P1	92,134	63.2	IncP	MOB_P	IncP-1	1.00	High
P2	56,781	61.8	IncA/C	MOB_F	IncA/C2	0.95	High
P3	48,221	60.5	IncF	MOB_F	IncFII	0.87	Medium
P4	34,109	62.1	IncN	MOB_V	IncN	0.72	Medium
P5	28,456	59.8	IncX	MOB_X	IncX1	0.65	Low
P6	21,903	64.0	IncF	MOB_F	IncFIB	0.58	Low
P7	15,662	58.9	Col	MOB_C	ColRNAI	0.41	Low
P8	9,887	55.6	–	–	–	0.22	No plasmid
P9	6,431	57.1	–	–	–	0.08	No plasmid
P10	4,993	58.3	–	–	–	0.02	No plasmid

Evaluation of PICCIS performance across different use cases

Sample	Niche	Species	Country	Type of file
MS14	Environment	B. contaminans	United States	.fasta
LMG23361	Environment	B. contaminans	Spain	.gbk
LMG23361fragmented	Environment	B. contaminans	Spain	.fasta
M1	Clinic	B. contaminans	Argentina	.fasta
M2	Clinic	B. contaminans	Argentina	.fastq.tar.gz
M3	Clinic	B. contaminans	Argentina	.fastq.tar.gz
CFSAN064036	Environment	Escherichia coli	United States	.fasta

PICCIS Score

Evaluation of PICCIS performance across different use cases

Detection heatmap by tool

Sample	SPAdes	MOB-recon	geNomad	Platon	PlasmidFinder
MS14	N/D	N/D	N/D	N/D	N/D
M1	N/D	N/D	N/D	N/D	N/D
LMG23361fragmented	N/D	N/D	N/D	N/D	N/D
M3_6	0	0	0	1	0
M2_5	0	0	1	1	0
M2_4	0	0	1	1	0
M3_5	1	0	0	1	0
M3_4	0	0	1	1	0
M3_3	0	0	1	1	0
M3_2	0	0	1	1	0
M2_3	0	0	1	1	0
M3_1	0	0	1	1	0
M2_2	0	0	0	1	1
CFSAN064036_4	N/D	1	0	0	1
LMG23361_4	N/D	0	1	1	0
CFSAN064036_3	N/D	1	0	0	1
LMG23361_3	N/D	0	1	1	0
LMG23361_2	N/D	1	1	1	0
LMG23361_1	N/D	1	1	1	0
CFSAN064036_2	N/D	1	1	1	1
CFSAN064036_1	N/D	1	1	1	1
M2_1	1	1	1	1	1

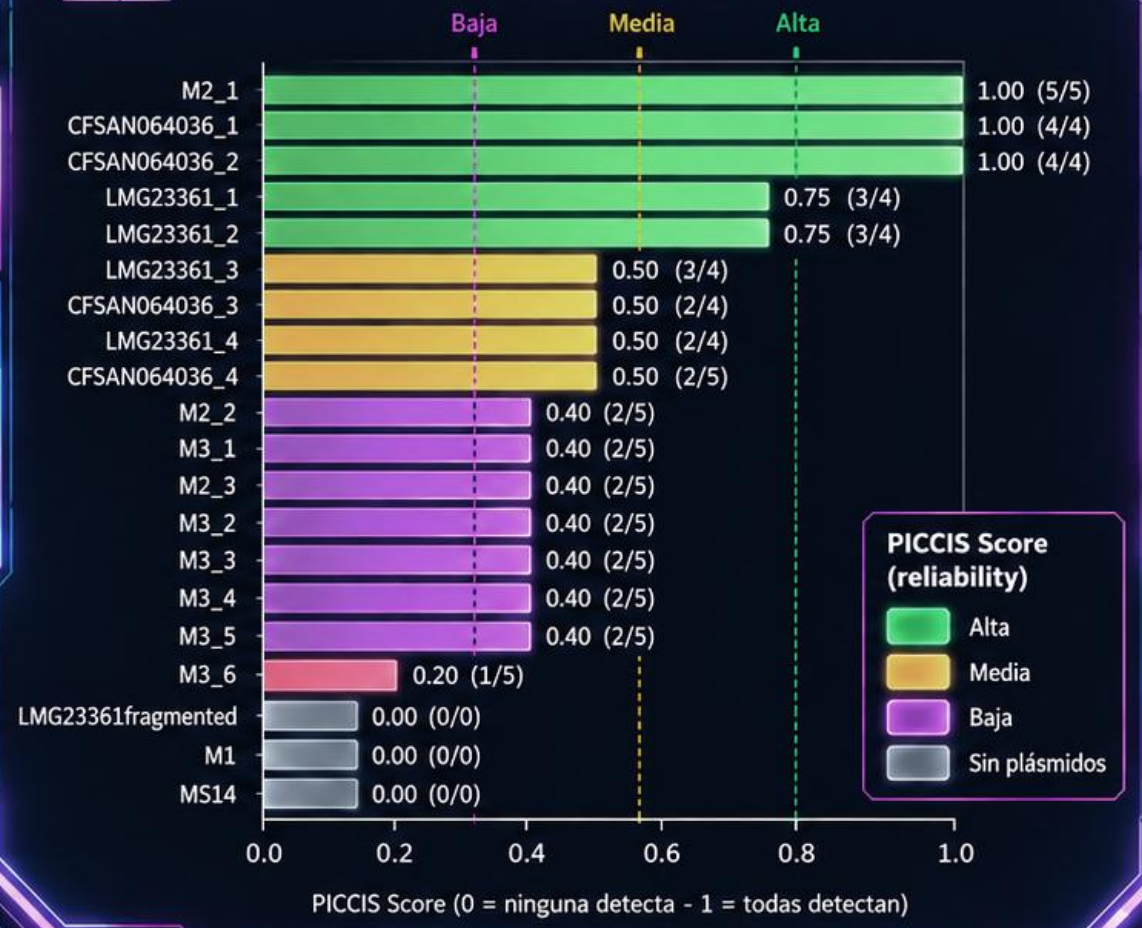
Legend

1 = Detected

= Not detected

1 N/D = Not applicable

PICCIS score by sample

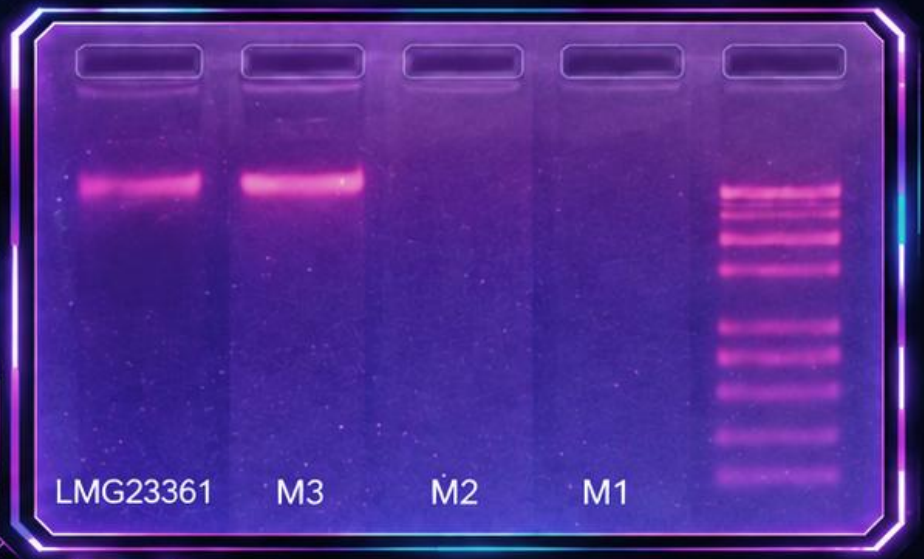


PICCIS Score

Case Study: In Silico Meets In Vitro

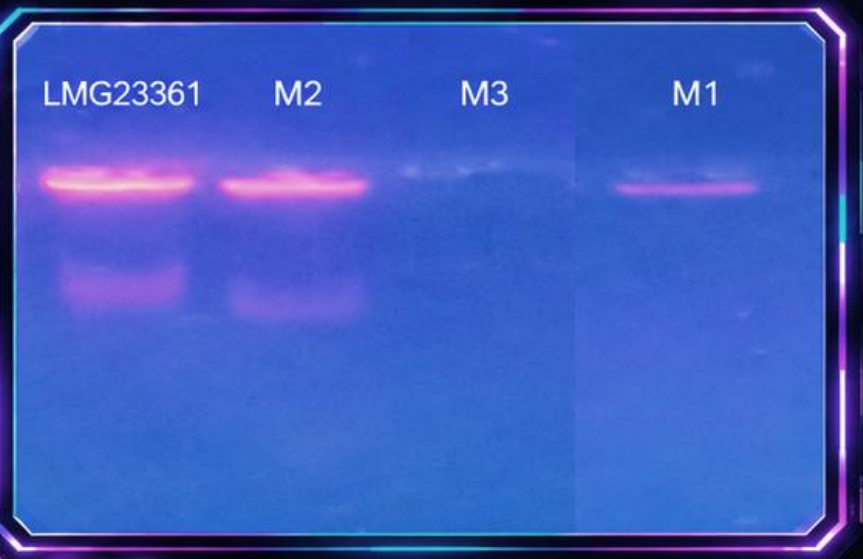
Clinical isolates of *Burkholderia contaminans* from Cystic Fibrosis (CF) patients.

A) Eckhardt



Agarose gel (viewed under UV light)

B) Miniprep



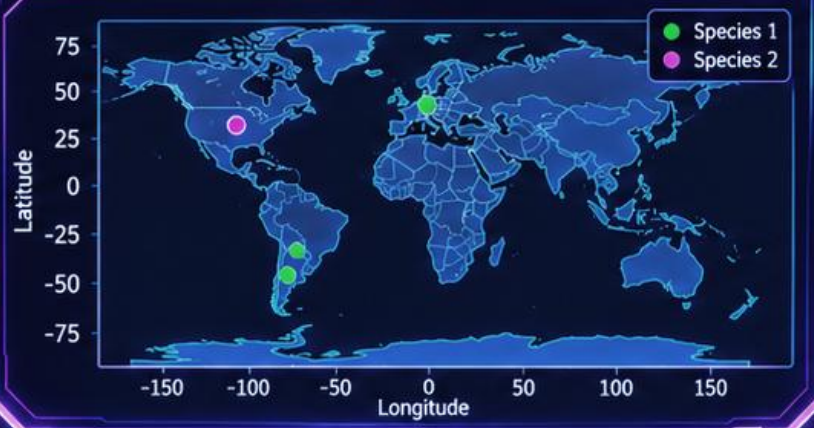
Agarose gel (viewed under UV light)

Key findings

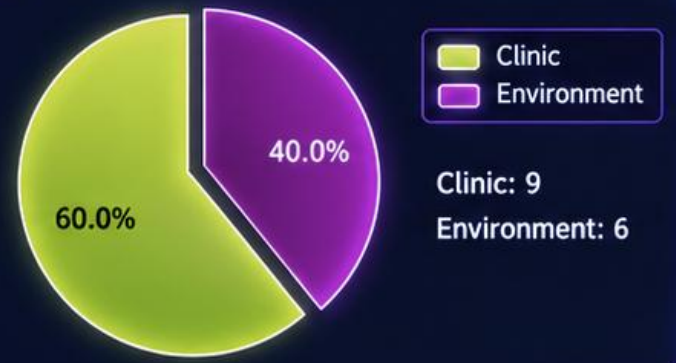
- **High PICCIS Score** predicted for isolates M2 and M3.
- **No plasmid predicted** for isolate 1.

Plasmid Characterization Features

Geographical Distribution of Plasmids by Species



Distribution by Environment



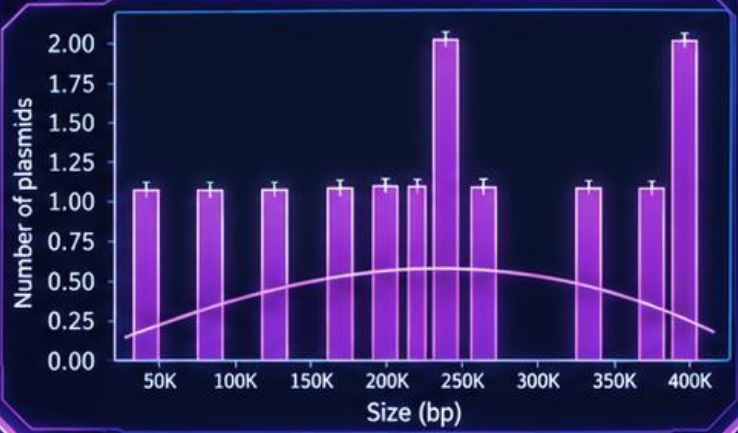
Plasmids by Species and Niche



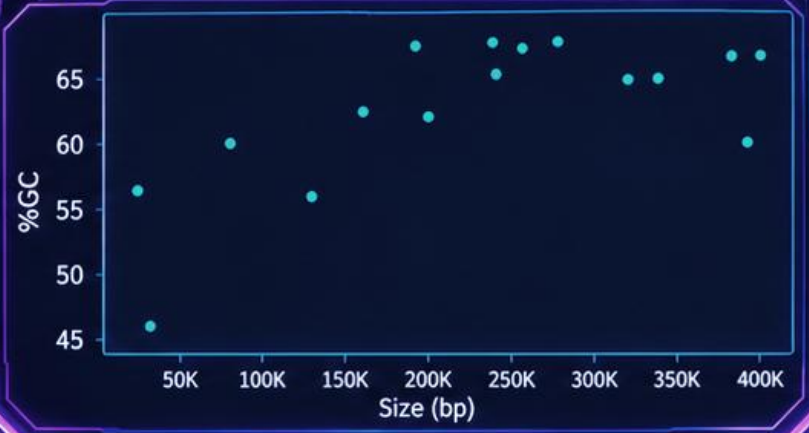
%GC Distribution by Environment



Size Distribution of Plasmids

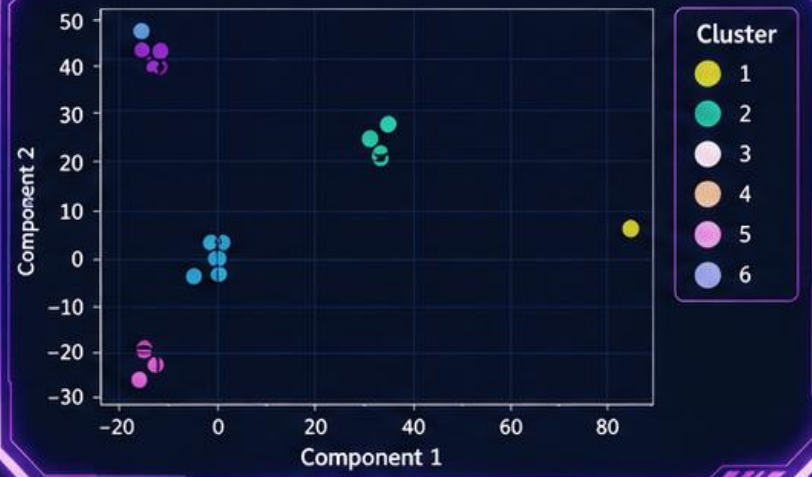


Size vs %GC

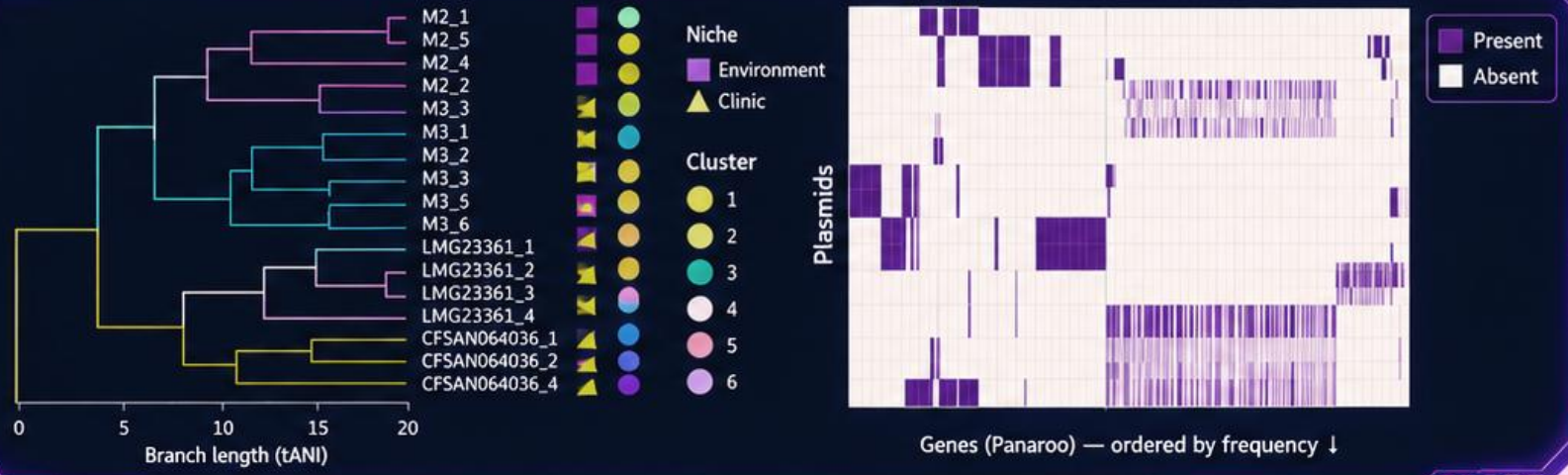


Plasmid Characterization and Genomic Analysis

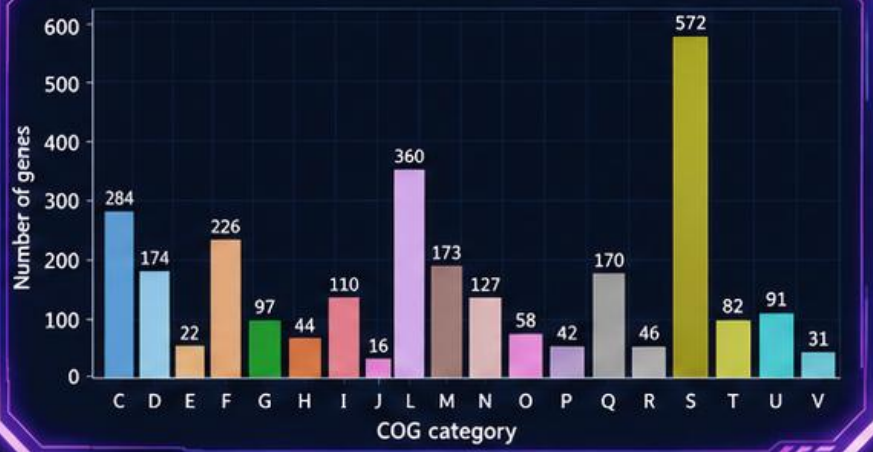
K-means (k = 6) – PCA + Unified Table (SVD)



Phylogenetic tree (tANI) and gene presence/absence



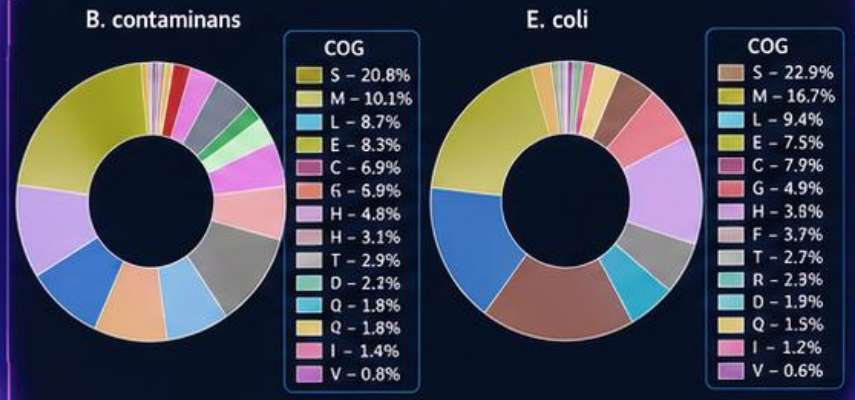
COG Functional Categories – All Plasmids



COG categories per plasmid



COG categories per species



Thank you for your attention.

Are there any questions?



GitHub: https://github.com/macarenaalga/PICCIS.git

DOI: 10.5281/zenodo.21358968

IBBM CONICET UNLP

CINDEFI CONICET UNLP